

ABIR SAHA

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Deep Learning · LLM · Explainable AI · Data Engineering · Cloud Computing · Database Systems · AI in Healthcare

EDUCATION

Southeast Missouri State University

Master of Science in Applied Computer Science

United States

Spring 2025 – Present

GPA: 4.0 / 4.0

- **Thesis:** Deep Learning–Driven EEG Seizure Prediction with Explainable AI Techniques
- Trained CNN+BiLSTM model on a benchmark EEG Seizure dataset (20 channels, 1280 time points), achieving 72.06% accuracy with ROC-AUC of 77%.
- Applied SHAP (Explainable AI) to analyze EEG channel contributions for global and local explanations; identified dominant channels CH2 and CH19.
- Uncovered error patterns in False Negatives, deepening model interpretability and diagnostic insights.
- Compared inference time of CNN+BiLSTM against baseline models including ViT-1, ViT-2, and Longformer.
- **Courses:** Advanced AI, Distributed Cloud Computing, Research Methods, Advanced Database Systems, Data Analytics, Cybersecurity, HCI, Thesis I & II

East West University

Bachelor of Science in Computer Science

Bangladesh

2015 – 2019

GPA: 3.06 / 4.0

- **Thesis:** Automatic Bengali Speech Recognition Using Probabilistic Language Model
- **Relevant Courses:** Machine Learning, Data Mining, Artificial Intelligence, Database Management Systems

PUBLICATIONS

- Saha, A., Pu, X., Ali, I., & Shuja, J. (2025, September). ML-Driven Loan Approval: Evaluating Predictive Models and Real-Time Deployment on the Cloud. In 3rd International Conference on AI, Blockchain, and IoT (AIBThings). IEEE.
- Saha, A., Saha, A., & Mittra, T. (2019, July). Performance measurements of machine learning approaches for prediction and diagnosis of chronic kidney disease (CKD). In Proceedings of the 7th International Conference on Computer and Communications Management (pp. 200–204).
- Sarowar, M. G., Jamal, A. A., Saha, A., & Saha, A. (2019, December). Performance evaluation of feature extraction and dimensionality reduction techniques on various ML classifiers. In IEEE 9th International Conference on Advanced Computing (IACC) (pp. 19–24). IEEE.

Under Review / In Progress:

- Can AI Detect Its Own Generated Peer Reviews? An Empirical Study
- A Review and Taxonomy of the Application of Open-source Machine Learning for EEG Data
- Computational Efficient EEG-Based Seizure Prediction Using Deep Learning with Explainable AI

PROFESSIONAL EXPERIENCE

Southeast Missouri State University – Dept. of Computer Science

Graduate Teaching Assistant

Cape Girardeau, MO

Fall 2024 – Spring 2025

- Assisted in teaching undergraduate CS courses; graded assignments and provided constructive feedback based on rubrics.
- Conducted research on student software engineering submissions to develop summer research on effective AI usage, guiding students to leverage AI tools productively rather than dependently.
- **Courses Supported:** Operating Systems (CS380), Software Engineering (CS445), Cybersecurity (CS501), Network Security (CS530), Distributed Cloud Computing (CS506), Web Programming & Security (IS245)

Southeast Missouri State University – Dept. of Computer Science

Research Assistant (Academic Engagement)

Cape Girardeau, MO

Summer 2025

Non-contractual, unpaid academic research under the supervision of Dr. Junaid Shuja

- Conducted a literature review on EEG-based studies utilizing machine learning methods.
- Led study on ML-driven loan approval systems with real-time AWS deployment; published in IEEE AIBThings 2025.

Frenclub

Software Engineer

Dhaka, Bangladesh

11/2021 – 11/2024

- Developed a delivery rider app streamlining order placement, tracking, and delivery workflows, improving user experience and operational performance by 25%.
- Maintained HR-related enterprise apps used by large companies, delivering faster and more organized tooling.
- Wrote clean, well-structured, and maintainable code following MVP, MVVM, and MVC architectures.

GuruKlub

Software Engineer

01/2021 – 11/2021

- Built and maintained learning features including quizzes, live exams, and level-up games to support B.C.S. exam preparation.
- Wrote clean, optimized, reusable code that improved app performance and smoothness for 10,000+ users.
- Implemented pixel-perfect Material Design responsive UI; maintained GitHub repository for collaborative workflow.

National Polytechnical Institute

Lecturer

Dhaka, Bangladesh

01/2020 – 12/2020

- Taught C programming and database management systems across two semesters.
- Managed grade sheets, assignment evaluations, and mid/final exam grading.
- Provided project guidance and feedback to students, fostering innovative thinking and identifying research gaps.

TECHNICAL SKILLS

- **Languages & Technologies :** Python, Dart, Java, Kotlin, C++, PHP, JavaScript, TypeScript
- **Frameworks & Tools:** TensorFlow, Flutter, MLflow, AWS (SageMaker, Lambda, S3), MySQL, LaTeX, Apache Tomcat, XAMPP, GitHub, MS Office, Confluence, Slack
- **Dev Environments:** VS Code, Android Studio, Anaconda Navigator, Eclipse, VirtualBox

TECHNICAL PROJECTS

Groq-Powered Portfolio Bot | Southeast Missouri State University | Python, AWS

- Created a personal AI agent using Groq LLaMA 3.3-70B that represents me online, answering questions about my background using LinkedIn and related documents as its knowledge source.
- Designed the agent to recognize its own limits and send instant Pushover notifications when it cannot answer, enabling iterative document expansion and continuous improvement.
- Deployed on Hugging Face after QA testing across corner cases and ambiguous queries.

Live: https://huggingface.co/spaces/sahaabir648/career_deploy_10

Loan Prediction Model with AWS SageMaker | Southeast Missouri State University | Python, AWS

- Balanced an imbalanced dataset; trained and compared Random Forest, Logistic Regression, XGBoost, and ANN models locally.
- Deployed the top-performing XGBoost model on AWS SageMaker and exposed it via AWS Lambda API, optimizing production cost.
- Published findings as a research paper at IEEE AIBThings 2025.

GitHub: <https://github.com/sahaabir917/Loan-prediction-model-with-aws-sagemaker>

Agentic Data Ingestion & Quality Pipeline | Southeast Missouri State University | Python, AWS, n8n, Databricks

- Built an agentic n8n workflow that retrieves invoices from Google Drive, extracts text, and uses Groq api to parse fields into Excel and upload to AWS S3, reducing manual data entry by 50%.
- Created a Databricks/Delta Lake Bronze-Silver-Gold medallion pipeline from three source systems with validation, cleaning, and Gold-layer role-based views.
- Developed a BI dashboard on 1.79B in sales data covering revenue trends, discount impact, peak-hour patterns, and product performance.

Co-PiiLOT Delivery AI GPS | Frenclub | Flutter, Android

- Flutter-based rider app supporting order acceptance, pickup, drop-off, and release with smart navigation, cutting delivery time by 30% and improving on-time delivery rates.

Play Store: https://play.google.com/store/apps/details?id=com.frenclub.pilot_public_rider

GuruKlub Learning App | Frenclub | Kotlin, RxJava, MVVM

- Built and launched MCQ, gamification, and Live Quiz features using Material Design and MVVM, increasing user retention by 35% across 10,000+ users; integrated billing for in-app game coins.
- Optimized app architecture with reusable components, reducing UI lag by 30% and improving interaction rates by 25%.

GitHub: <https://github.com/sahaabir917/Guruklub>

Play Store: <https://play.google.com/store/apps/details?id=com.gmpire.guruklub>

Rajendra College Admission & Payment App | Dynamic Host BD | Flutter, GetX

- Designed a scalable database architecture for National University admission workflows.
- Integrated SSL payment gateway (including Google Pay), reducing manual payment errors by ~30% and improving transaction reliability.

GitHub: <https://github.com/sahaabir917/Rajendra-College-Faridpur>

Play Store: <https://play.google.com/store/apps/details?id=com.student.payment>

Award

MSACS Outstanding Student Award (2025–2026) from Southeast Missouri State University

CERTIFICATIONS

- **Developing Explainable AI (XAI)** | Duke University, Coursera |
- Verify: <https://www.coursera.org/account/accomplishments/verify/PDYP10JKLMLX>

LEADERSHIP & SERVICE

- President, CS Researchers Club | September 2024 – Present | Southeast Missouri State University
- Member, AI Club | March 2024 – Present | Southeast Missouri State University

REFERENCES

- Dr. Junaid Shuja, Assistant Professor, Southeast Missouri State University, USA
- Dr. Ihsan Ali, Assistant Professor, Southeast Missouri State University, USA
- Dr. Mario Guimaraes, Professor & Chair of Computer Science, Southeast Missouri State University, USA